

VideoSQL: Ultra Long Video Understanding by Agentic SQL Queries with Trainable Skills

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Abstract

and is merged if it turns a failure into a success or maintains a success. However, in many cases, the two traces are completely independent (with no shared prefix beyond the prompt), or the divergence point does not affect the result.

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1. Introduction

Ultra long video understanding (ULVU)

2. Related Works

2.1. Ultra Long Video Understanding

2.2. Skill Training

Training model weights, especially with reinforcement learning (RL), is expensive, slow, and not portable (you would need to host and run this specific model, which might perform poorly on general tasks). Many recent works began to train the agent's Skill document instead of model weights, which can be seen as a form of prompt optimization.

Training-free GRPO [1] uses an approach similar to the original gradient-based GRPO [7]. An initial experience and a policy model are used to roll out a batch of traces, and another LLM computes the semantic advantage of each rollout, analogous to the numeric advantage in the original GRPO. These semantic advantages are used to update the experience. **Trace2Skill** [6] uses parallel rollouts in batches to collect many traces and generate patches for successful and failed traces. These patches are then merged into the final Skill in sequence. Compared to training-free GRPO, this can be seen as a form of offline reinforcement learning. **SkillOpt** [10] treats Skill optimization similarly to gradient descent, where the editing budget is analogous to the learning rate, the gradient direction is analogous to the editing direction, etc. It also learns a meta-skill to track what kind of edit is helpful using a held-out validation set. **SkillCAT** [3] uses DPO-style training. It samples two paired rollouts: a successful one (r^+) and a failed one (r^-). It then analyzes the two traces, finds the point where they diverge, and generates a patch contrastively. The patch is then verified to see whether it helps the agent, similar to the gate in SkillOpt,

3. VideoSQL

3.1. Index Building

We used the same index sturcture as MAGIC-Video [5], which includes video captions, entities, visual clips, event chain, etc. Readers can referenced to MAGIC-Video for more details. However, the original pipeline is complex and involving many slow stages like OpenIE. We modified the indexing process such that an VLM (Qwen 3.8) to directly output these information based on the input.

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3.2. SQL-based queries

MAGIC-Video uses a Personalized Page Rank (PPR) to retrieval nodes in the graph by a query generated by the LLM, which the agent has no control over the relations between nodes. For example, if the agents want to find who ate the apple yesterday noon, it likely issues a query like "person ate apple yesterday noon" and wish the PPR returns relevant nodes. Instead, we formulate this using a relational database. Given the same indecies sturcture as MAGIC-Video, the agent can write a SQL query like in Listing 1. It can also handle complex queries like ""

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Listing 1. An example SQL query for finding who ate the apple within a given time window.

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SELECT DISTINCT subject
FROM semantic_triples
WHERE LOWER(predicate) IN ('eat', 'eats'
, 'ate', 'eating')
AND LOWER(object) IN ('apple', 'the_
apple', 'an_apple')
AND snapshot_s >= $start_s
AND snapshot_s < $end_s;
```

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3.3. Trainable Skills

4. Evaluations

4.1. Datasets

We used two most popular ULVU datasets in the paper: EgoLifeQA [9] and MM-Lifelong [2]. EgoLifeQA uses multiple-choice questions (MCQ) while MM-Lifelong uses open QA. EgoLifeQA contains one week of recording in the same setting while MM-Lifelong has 3 different settings (day, week, and month) each with its own video. The MM-Lifelong week subset has the same video as EgoLifeQA. We argue that MCQ does not reflect realistic usage since most questions asked by users in real world are open. Additionally, agents can brute-force verify each option to artificially boost accuracy. We therefore reformulated EgoLifeQA into open QA questions (Open-EgoLifeQA) similar to MM-Lifelong and evaluated both using LLM-as-judge for semantic comparison between the ground-truth and predicted answers. To compare with prior work, we used the Jack view from EgoLifeQA. We evaluated our un-trained VideoSQL on both Open-EgoLifeQA and MM-Lifelong validation set.

To train the VideoSQL Skill, we need to select a independent training set. Ego-R1 Data [8] is a training set build on top of EgoLifeQA and authors of Ego-R1 trained their method on this dataset and then tested on EgoLifeQA. MM-Lifelong also has a training split as well. However, these training sets overlap with their corresponding validation/testing sets at the video level. While no questions are leaked, we argue that sharing videos could still inflate performance, or at least fail to test the model's ability for cross-domain generalization. To address these, we trained out Skill on MM-Lifelong day-long and week-long subset and evaluated on the most challenging MM-Lifelong month-long subset.

4.2. Baselines

We selected two baselines for the training free harness, MAGIC-Video [5] and GROVE [4], all using GPT-5.6-luna as policy model. This will be used to compare with our harness with untrained skills. We also selected Trace2Skill [6] and SkillOpt [10] for our skill training methods, both build on top of the same VideoSQL harness.

References

- [1] [2510.08191] Training-Free Group Relative Policy Optimization. <https://arxiv.org/abs/2510.08191>. 1
- [2] Guo Chen, Lidong Lu, Yicheng Liu, Liangrui Dong, Lidong Zou, Jixin Lv, Zhenquan Li, Xinyi Mao, Baoqi Pei, Shihao Wang, Zhiqi Li, Karan Sapra, Fuxiao Liu, Yindong Zheng, Yifei Huang, Limin Wang, Zhiding Yu, Andrew Tao, Guilin Liu, and Tong Lu. Towards Multimodal

Lifelong Understanding: A Dataset and Agentic Baseline. <https://arxiv.org/abs/2603.05484v1>, 2026. 2

- [3] Kunfeng Chen, Qihuang Zhong, Juhua Liu, and Bo Du. SkillCAT: Contrastive, Assessment-Augmented and Topology-AwareSkill Self-Evolution for LLM Agents. <https://arxiv.org/abs/2606.13317v2>, 2026. 1
- [4] Sitong Gong, Caixin Kang, Tianyu Yan, Guo Chen, Bo Zheng, Kaipeng Zhang, Yunzhi Zhuge, Xiang Ruan, Huchuan Lu, and Yifei Huang. GROVE: Growing and Reasoning over Temporally Stratified Memory from Streaming Video Experience, 2026. 2
- [5] Jiazheng Li, Chi-Hao Wu, Yunze Liu, Kaize Ding, Jundong Li, and Chuxu Zhang. Bridging Modalities, Spanning Time: Structured Memory for Ultra-Long Agentic Video Reasoning, 2026. 1, 2
- [6] Jingwei Ni, Yihao Liu, Xinpeng Liu, Yutao Sun, Mengyu Zhou, Pengyu Cheng, Dexin Wang, Erchao Zhao, Xiaoxi Jiang, and Guanjin Jiang. Trace2Skill: Distill Trajectory-Local Lessons into Transferable Agent Skills, 2026. 1, 2
- [7] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models, 2024. 1
- [8] Shulin Tian, Ruiqi Wang, Hongming Guo, Penghao Wu, Yuhao Dong, Xiuying Wang, Jingkang Yang, Hao Zhang, Hongyuan Zhu, and Ziwei Liu. Ego-R1: Chain-of-Tool-Thought for Ultra-Long Egocentric Video Reasoning, 2025. 2
- [9] Jingkang Yang, Shuai Liu, Hongming Guo, Yuhao Dong, Xiamengwei Zhang, Sicheng Zhang, Pengyun Wang, Zitang Zhou, Binzhu Xie, Ziyue Wang, Bei Ouyang, Zhengyu Lin, Marco Cominelli, Zhongang Cai, Yuanhan Zhang, Peiyuan Zhang, Fangzhou Hong, Joerg Widmer, Francesco Gringoli, Lei Yang, Bo Li, and Ziwei Liu. EgoLife: Towards Egocentric Life Assistant, 2026. 2
- [10] Yifan Yang, Ziyang Gong, Weiquan Huang, Qihao Yang, Ziwei Zhou, Zisu Huang, Yan Li, Xuemei Gao, Qi Dai, Bei Liu, Kai Qiu, Yuqing Yang, Dongdong Chen, Xue Yang, and Chong Luo. SkillOpt: Executive Strategy for Self-Evolving Agent Skills, 2026. 1, 2